Deconstructing Credit Risk: A Bayesian Structural Analysis of Lending Club Data

From Black-Box Scores to Causal Interpretability

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Context & Research Objective

The Problem: P2P Lending Risk

- Lending Club operates as a marketplace connecting borrowers and investors.
- Unlike traditional banking, risk assessment relies heavily on engineered features.

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- In regulated finance, understanding *why* a loan is classified as risky is mandatory.

Objective: Use **Bayesian Networks (BN)** to audit structural logic and uncover hidden causal dependencies.

Data Preprocessing & Discretization

- **Dataset:** Lending Club Loan Data (2007-2018) reduced to $\sim$ 50,000 cases.

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- **Dataset:** Lending Club Loan Data (2007-2018) reduced to $\sim$ 50,000 cases.
- **Preprocessing Strategy:**
  - *Feature Subset Selection (FSS):* Reduced from 151 to 29 variables using Mutual Information (MI) and correlation filters.
  - *Discretization:* Quantile-based discretization (5 bins) for continuous variables (e.g., `annual_inc`, `dti`).

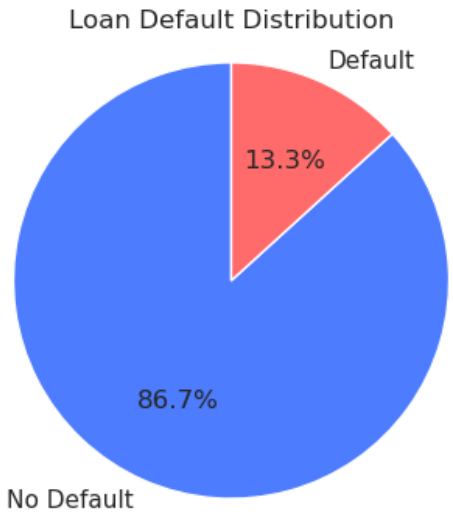

Figure: Loan Status Distribution

Why Discretization?

Crucial for robust structure learning to prevent outliers from skewing distributions.

Methodology: Structure Learning Approaches

We evaluated two distinct paradigms to learn the Directed Acyclic Graph (DAG):

1. **Search-and-Score Approach (Selected)**
   - **Algorithm:** Hill Climbing (HC) with Tabu Search optimization.
   - **Scoring Metrics Evaluated:**
     - **K2:** Standard Bayesian score.
     - **AIC:** Akaike Information Criterion.
     - **BIC:** Bayesian Information Criterion (Selected for its heavy penalty on complexity).
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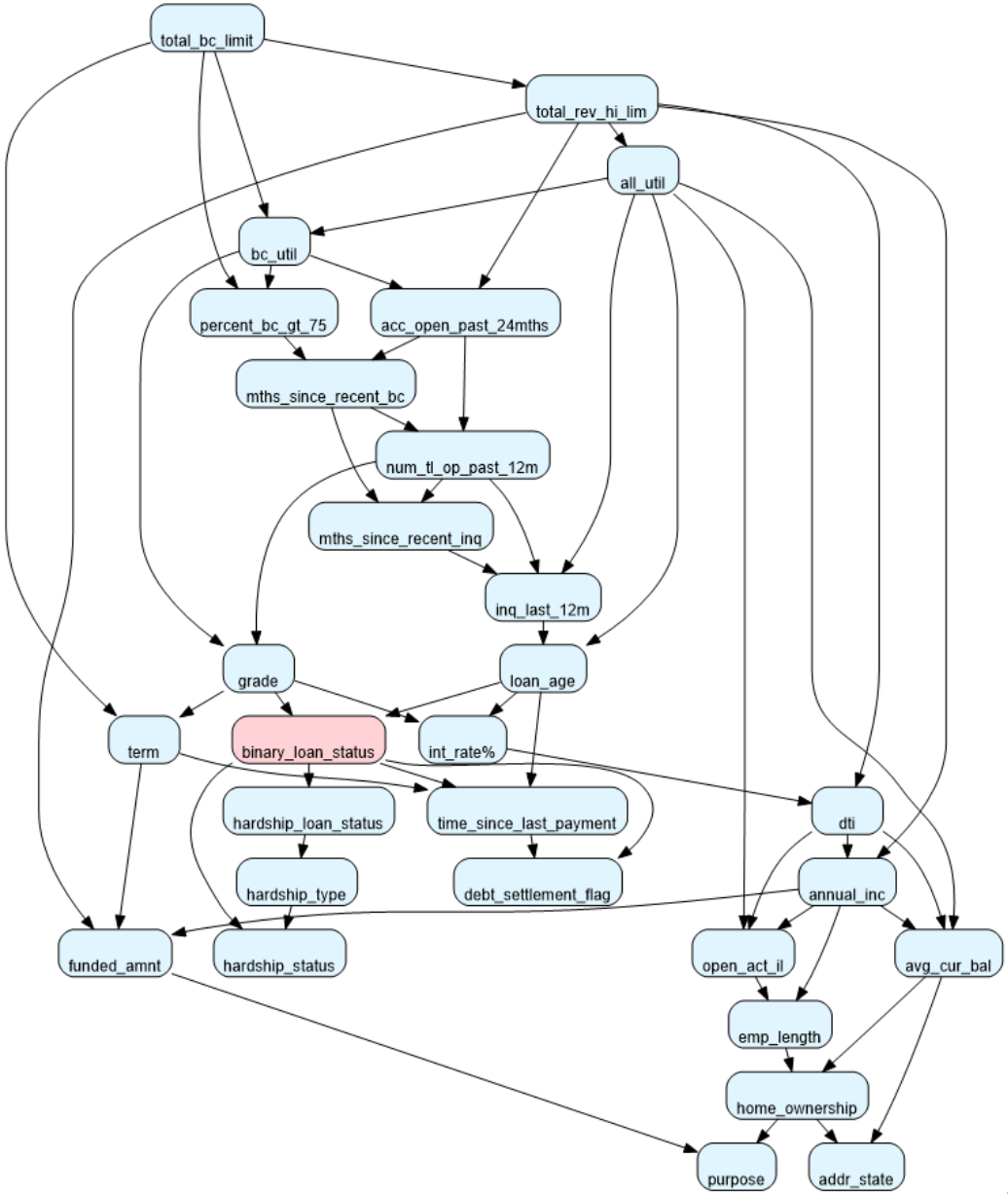
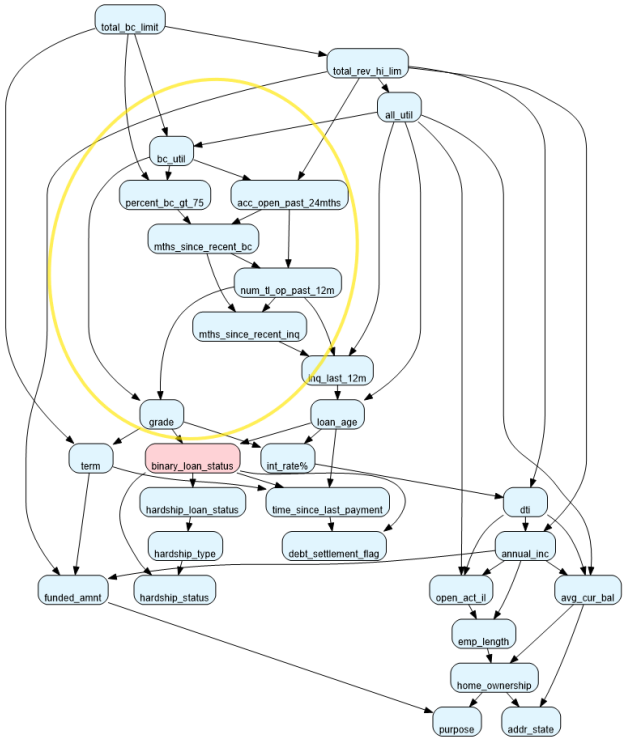
2. **Constraint-Based Approach (Discarded)**
   - **Algorithm:** PC Algorithm.
   - **Mechanism:** Uses conditional independence tests (χ^2 / G^2) starting from a fully connected graph.

Initial Model: Full HC Algorithm

The initial HC model revealed a centralized structure dominated by `grade` and `loan_age` variable.

- **The Hub:** they act as the parents for `int_rate` and the direct predictor of `loan_status`.
- **Information Funnel:** Almost all financial history flows into `grade` and `loan_age`.

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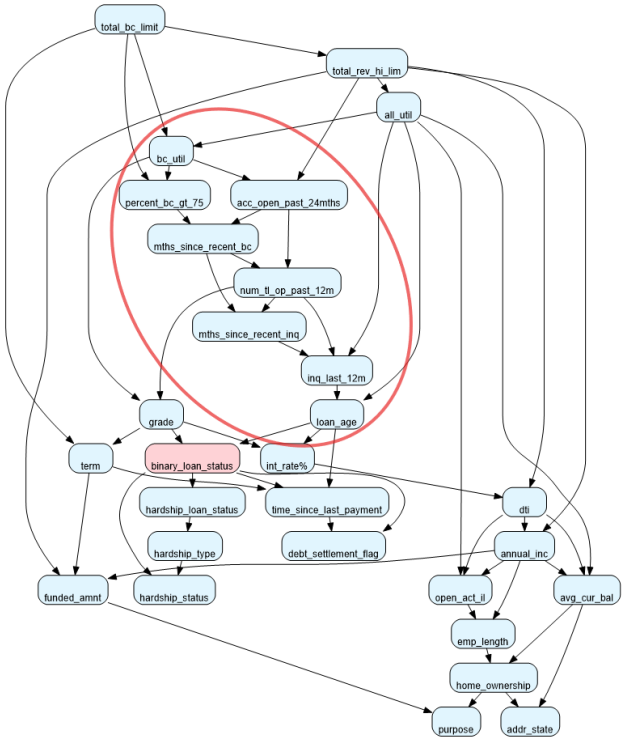
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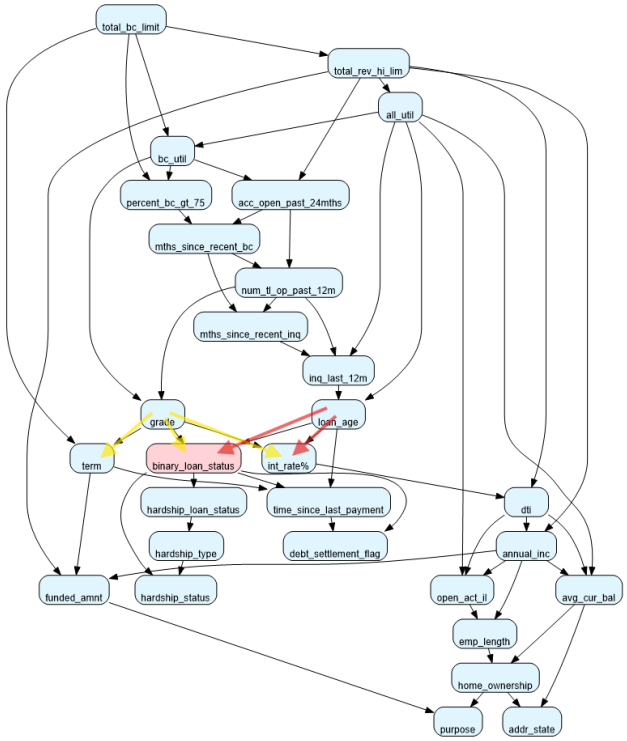
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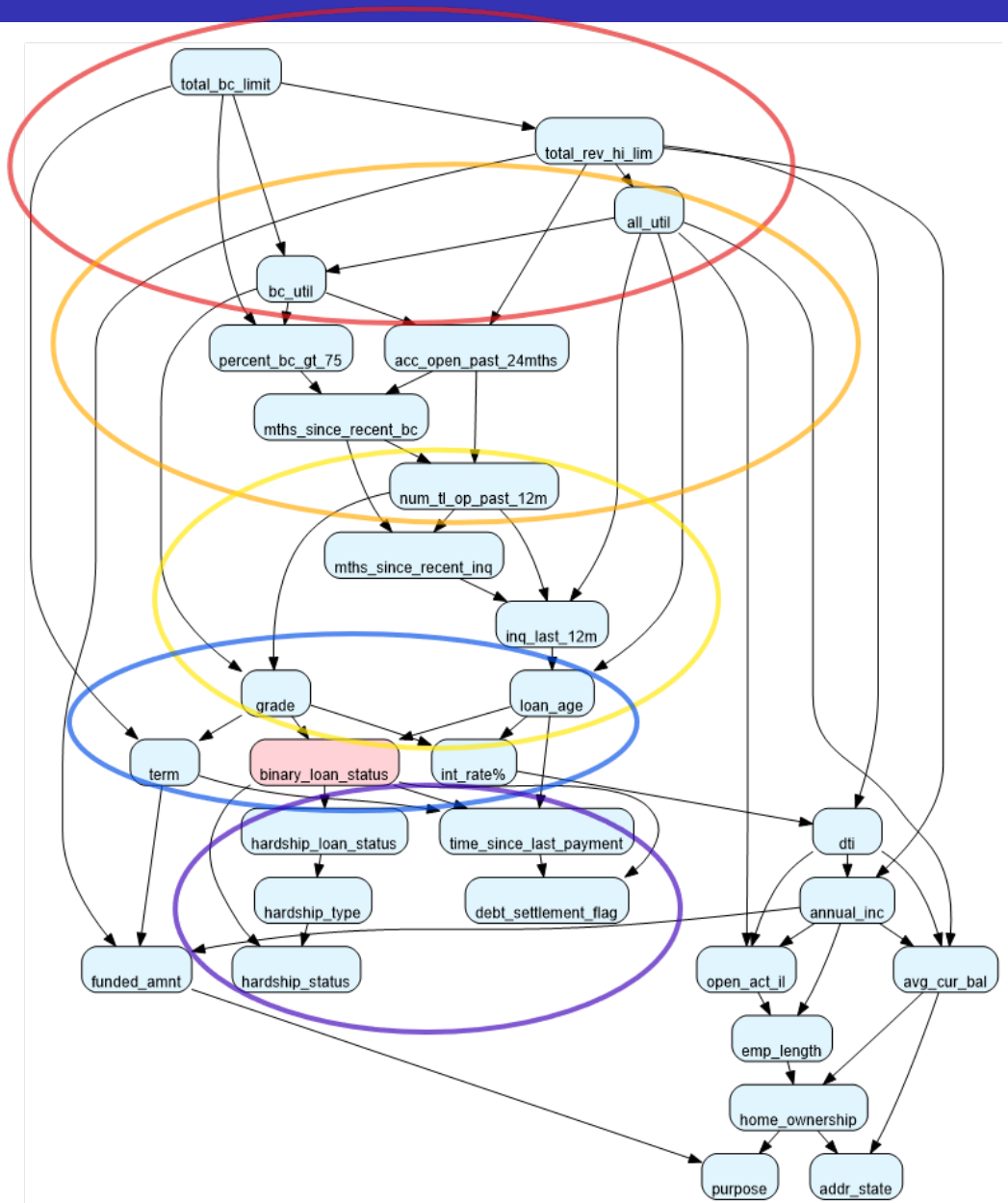
Deep Dive: The Behavioral Cascade

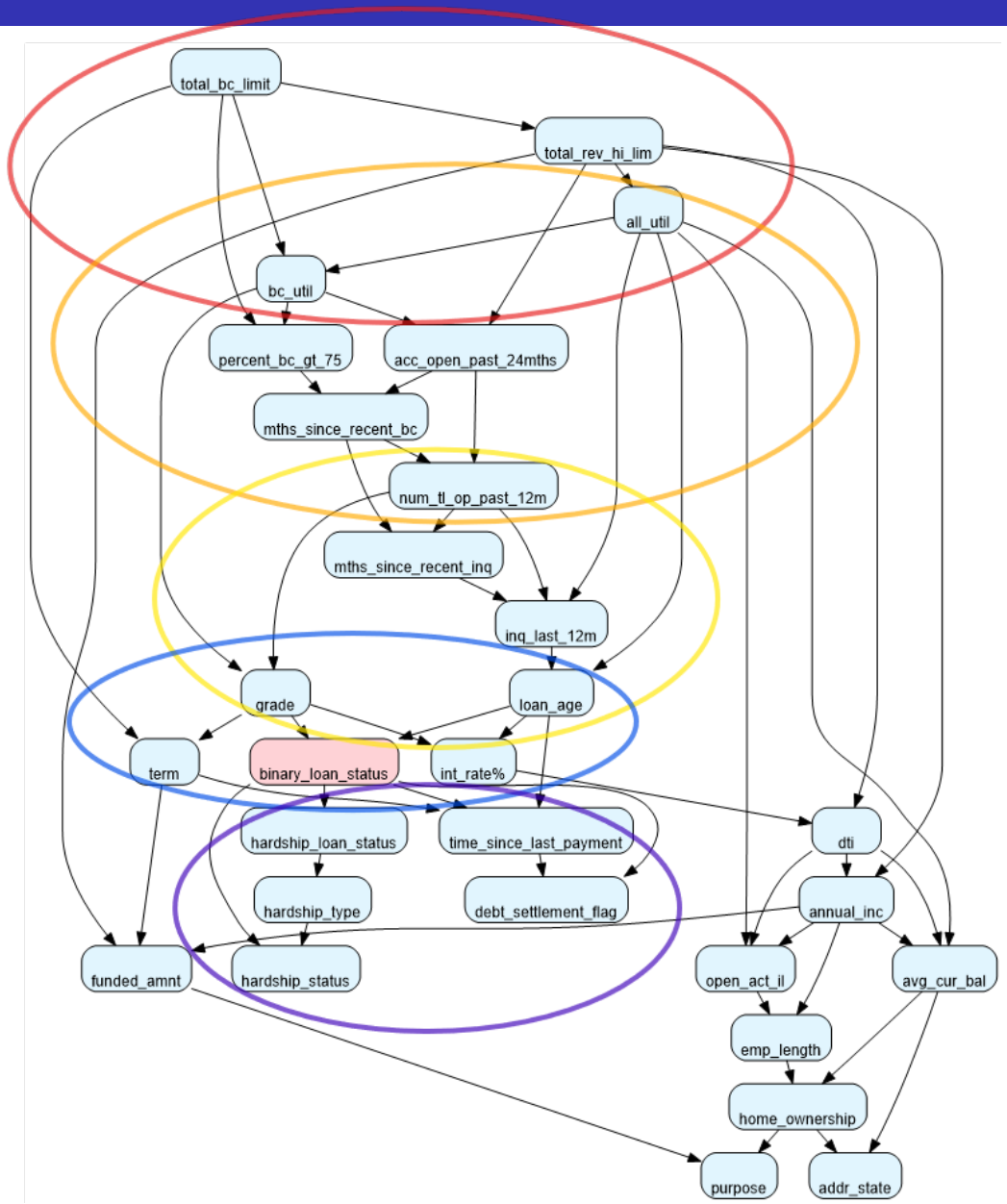
The network reveals a clear causal narrative driving the risk of default.

- **1. Capacity (Red):** Credit limits like `total_bc_limit` define the initial financial baseline.
- **2. Saturation (Orange):** High utilization (`bc_util`, `acc_open`,) flows downwards as a sign of financial stress.
- **3. Desperation (Yellow):** Saturation drives the search for new credit (`num_since_rcnt_inq`, `inq_last_12m`).



The Outcome: Incomplete model

- **4. `loan_status` brothers (blue):** By contract design, default probability depends directly on the loan `term` and the `int_rate`.
- **5. Consequences (purple):** Downstream nodes like `hardship_loan_status` and `debt_settlement_flag` are **children** of `loan_status`; they are the symptoms.
- **Socioeconomic Exclusion:** Variables like `annual_inc`, `dti`, and `home_ownership` are structurally separated from the outcome.



Inferences, Variable Elimination and Markov Blankets

Variable elimination: $P(\text{loan_status} \mid \text{loan_age}, \text{grade})$

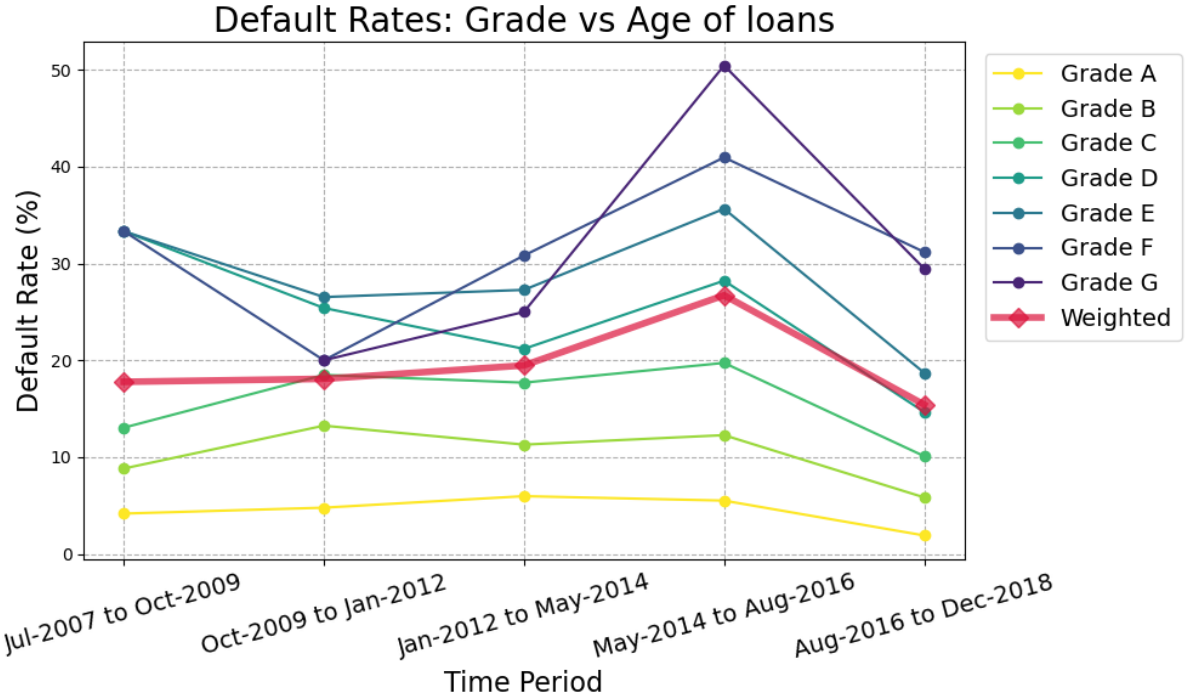


Table 1: Default Rate by Grade and Time Period

Grade	Jul 07–09	Oct 09–12	Jan 12–14	May 14–16	Aug 16–18	Marginal P(Def)
A	4.17%	4.78%	5.97%	5.51%	1.91%	**3.72%**
B	8.82%	13.25%	11.29%	12.27%	5.84%	**9.08%**
C	13.04%	18.49%	17.69%	19.73%	10.10%	**14.99%**
D	33.33%	25.40%	21.18%	28.22%	14.66%	**21.18%**
E	33.33%	26.53%	27.27%	35.64%	18.71%	**26.94%**
F	33.33%	20.00%	30.82%	40.95%	31.18%	**35.09%**
G	–	20.00%	25.00%	50.44%	29.47%	**37.60%**

Inference Validation (A): The "Funnel" Effect

Hypothesis: If `grade` is a sufficient statistic, adding `dti` info should not change the probability of default.

$$P(Default|Grade) \approx P(Default|Grade, DTI = High)$$

Scenario	Base P(No Def)	With DTI=High	Impact (Δ)
Grade A (Low Risk)	0.9628	0.9631	**+0.0003**
Grade B	0.9092	0.9099	**+0.0007**
Grade C	0.8501	0.8513	**+0.0012**
Grade D	0.7882	0.7920	**+0.0038**
Grade E	0.7306	0.7317	**+0.0011**
Grade F (High Risk)	0.6491	0.6498	**+0.0007**

Conclusion: Hypothesis **Confirmed**. The variation is **negligible** ($< 0.38\%$) across all profiles. The `grade` node effectively encapsulates risk, rendering `dti` conditionally independent.

Inference Validation (B): Sensitivity to Purpose

Hypothesis: Investment purposes (Education/Energy) should be safer than Consumption purposes (Wedding/Car).

$$P(\text{Success} \mid \text{Amt} = X, \text{Purp} = \text{Inv}) > P(\text{Success} \mid \text{Amt} = X, \text{Purp} = \text{Cons})$$

Loan Amount	**Best Purpose**	**Worst Purpose**	**Gap**
1k - 7k (Min)	Educational	Car	0.27%
7k - 10.5k	Home Improv.	Educational	0.13%
10.5k - 15.1k	Educational	Wedding	0.81%
15.1k - 23.3k	Renewable Energy	Wedding/Car	0.82%
23.3k - 40k (Max)	Renewable Energy	Debt Consolid.	0.12%

Inference Validation (C): Sensitivity to Purpose

Loan Amount	Best Purpose	Worst Purpose	Gap
1k - 7k (Min)	**Educational**	Car	1.27%
7k - 10.5k	**Home Improv.**	**Educational**	0.58%
10.5k - 15.1k	**Educational**	Wedding	1.81%
15.1k - 23.3k	**Renewable Energy**	Wedding/Car	2.82%
23.3k - 40k (Max)	**Renewable Energy**	Debt Consolid.	1.32%

Interpretation:

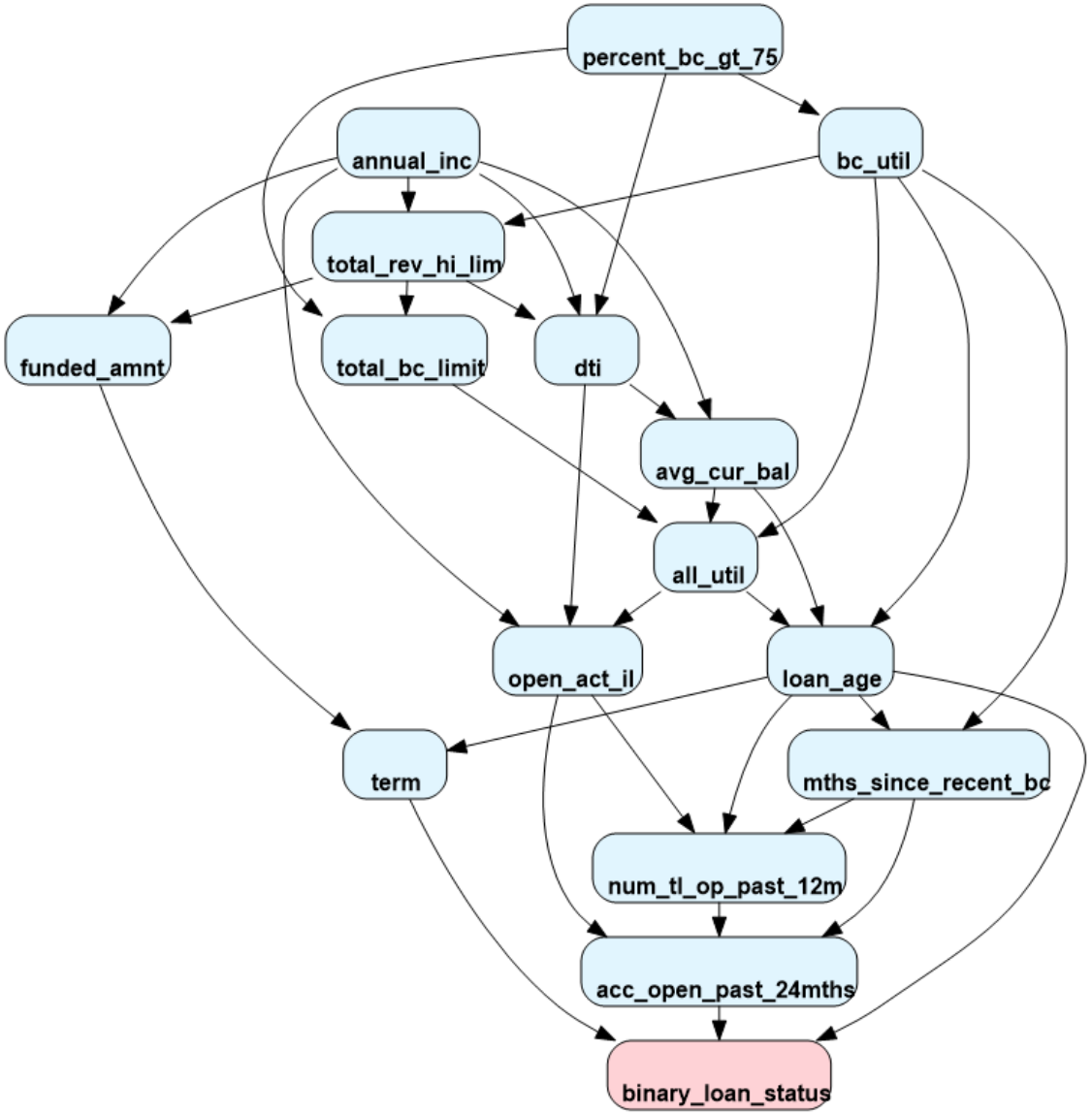
- **Low Sensitivity:** The predictive gap is small ($< 2.82\%$), meaning purpose is largely blocked by other factors.
- **Qualitative Trend:** "**Investment**" logic holds: *Renewable Energy* or *Education* are safer than *Weding* or *Car*.
  Debt Consolidation is riskiest for very large loans (distress signal).

The Refined Model: "Sanitizing" the Network

To uncover the underlying causal mechanisms, we applied **Expert Constraints**:

1. **Blacklisting "Grade":** Removed explicitly to break the information bottleneck and force the network to learn from raw data.
2. **Leakage Removal:** Eliminated post-default variables (e.g., `hardship_status`, `debt_settlement`) that were artificially inflating accuracy.
3. **Temporal Order:** Forbidden arcs from Target $\rightarrow$ Origination Features (e.g., Term, Amount).

Final Results: The True Drivers of Risk



New Markov Blanket:
Without *Grade*, risk depends on:

1. **Term:** Structural exposure (60m vs 36m).
2. **Volatility:** `acc_open_past_24mths`.
3. **Time:** `loan_age`.

Conclusions

1. **Score Efficiency:** The Lending Club `grade` is extremely efficient, effectively blocking raw variables. It works, but it's a black box.
2. **Algorithmic Suitability:** In high-dimensional discrete datasets, **Search-and-Score (HC)** outperforms **Constraint-Based (PC)** methods, which suffer from false independence discovery.
3. **Structural Auditing:** Bayesian Networks successfully identified:
   - Temporal and causal effects.
   - The "Solvency Tree" (Income $\rightarrow$ Limits $\rightarrow$ Utilization).
   - The critical role of **credit-seeking behavior** over static financial ratios.